



TRIOS CYBER

LABORATORY PRACTICE 6

Failed Authentication Attempts and Log Analysis

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1. Laboratory Account Verification

The active Kali Linux laboratory account was verified with the `whoami` command before generating authentication events. The system identified the current user as `matthew`, establishing the account used for the controlled failed-login activity and providing a clear reference for later log correlation.

```
(matthew@kali-mv)-[~]
$ whoami
matthew
(matthew@kali-mv)-[~]
```

2. Safe Failed Authentication Attempts

Three controlled authentication failures were generated against the local `matthew` account using `su - matthew`. An intentionally incorrect password was entered for each attempt, and the terminal returned “Fallo de autenticación” after every test. The number of attempts was kept low so the required events were created without repeatedly testing the account or risking an account lockout.

```
(matthew@kali-mv)-[~]
$ su - matthew
Contraseña:
su: Fallo de autenticación

(matthew@kali-mv)-[~]
$ su - matthew
Contraseña:
su: Fallo de autenticación

(matthew@kali-mv)-[~]
$ su - matthew
Contraseña:
su: Fallo de autenticación
```

3. Authentication Event Search with journalctl

The resulting system events were searched with `journalctl` using a ten-minute time window and filters for authentication failure, FAILED SU, and `pam_unix`. The journal contained `pam_unix(su-l:auth): authentication failure` entries and corresponding FAILED SU records for the `matthew` account. The events also identified the local terminal session as `pts/0`, allowing the generated failures to be matched to the manual test activity.

```
(matthew@kali-mv)-[~]
$ sudo journalctl --since "10 minutes ago" | grep -Ei "authentication failure|FAILED SU|pam_unix"
[sudo] contraseña para matthew:
sep 07 22:16:41 kali-mv lightdm[903]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=123) by (uid=0)
sep 07 22:16:41 kali-mv (systemd)[926]: pam_unix(systemd-user:session): session opened for user lightdm(uid=123) by lightdm(uid=0)
sep 07 22:16:44 kali-mv lightdm[903]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=123) by (uid=0)
sep 07 22:17:01 kali-mv CRON[1309]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
sep 07 22:17:01 kali-mv CRON[1309]: pam_unix(cron:session): session closed for user root
sep 07 22:17:05 kali-mv lightdm[903]: pam_unix(lightdm-greeter:session): session closed for user lightdm
sep 07 22:17:05 kali-mv lightdm[903]: pam_unix(lightdm-greeter:session): session closed for user lightdm
sep 07 22:17:05 kali-mv lightdm[1314]: pam_unix(lightdm:session): session opened for user matthew(uid=1000) by matthew(uid=0)
sep 07 22:17:05 kali-mv (systemd)[1338]: pam_unix(systemd-user:session): session opened for user matthew(uid=1000) by matthew(uid=0)
sep 07 22:17:16 kali-mv (sd-pam)[932]: pam_unix(systemd-user:session): session closed for user lightdm
sep 07 22:19:22 kali-mv su[3669]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=/dev/pts/0 ruser=matthew rhost= user=matthew
sep 07 22:19:24 kali-mv su[3669]: FAILED SU (to matthew) matthew on pts/0
sep 07 22:19:45 kali-mv su[3873]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=/dev/pts/0 ruser=matthew rhost= user=matthew
sep 07 22:19:47 kali-mv su[3873]: FAILED SU (to matthew) matthew on pts/0
sep 07 22:19:51 kali-mv su[3925]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=/dev/pts/0 ruser=matthew rhost= user=matthew
sep 07 22:19:54 kali-mv su[3925]: FAILED SU (to matthew) matthew on pts/0
```

4. Recent Authentication Event Review

A narrower five-minute journal review was then performed to isolate the most recent authentication-related activity. The filtered output showed the failed authentication event associated with matthew and provided a concise view of the event near the time of the test. A second review filtered su-related journal messages and displayed the surrounding PAM and sudo session activity, providing additional context for the authentication timeline.

```
(matthew@kali-mv)-[~]
$ sudo journalctl --since "5 minutes ago" | grep -i "authentication"
sep 07 22:19:51 kali-mv su[3925]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=/dev/pts/0 ruser=matthew rhost= user=matthew
sep 07 22:23:26 kali-mv sudo[5714]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/grep -Ei 'authentication failure|FAILED SU|pam_unix' /var/log/auth.log

(matthew@kali-mv)-[~]
$ sudo journalctl --since "5 minutes ago" | grep -i "su"
sep 07 22:21:41 kali-mv sudo[4803]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/journalctl --since '10 minutes ago'
sep 07 22:21:41 kali-mv sudo[4803]: pam_unix(sudo:session): session opened for user root(uid=0) by matthew(uid=1000)
sep 07 22:21:41 kali-mv sudo[4803]: pam_unix(sudo:session): session closed for user root
sep 07 22:23:26 kali-mv sudo[5714]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/grep -Ei 'authentication failure|FAILED SU|pam_unix' /var/log/auth.log
sep 07 22:23:26 kali-mv sudo[5714]: pam_unix(sudo:session): session opened for user root(uid=0) by matthew(uid=1000)
sep 07 22:23:26 kali-mv sudo[5714]: pam_unix(sudo:session): session closed for user root
sep 07 22:24:47 kali-mv sudo[6363]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/journalctl --since '5 minutes ago'
sep 07 22:24:47 kali-mv sudo[6363]: pam_unix(sudo:session): session opened for user root(uid=0) by matthew(uid=1000)
sep 07 22:24:47 kali-mv sudo[6363]: pam_unix(sudo:session): session closed for user root
sep 07 22:25:24 kali-mv sudo[6671]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/journalctl --since '5 minutes ago'
sep 07 22:25:24 kali-mv sudo[6671]: pam_unix(sudo:session): session opened for user root(uid=0) by matthew(uid=1000)
```

5. Authentication Event Correlation

The command output and system journal were compared to confirm that the deliberately generated failures were recorded by the operating system. The account name, authentication module, terminal session, and event timing visible in journalctl provided enough context to associate the log entries with the three failed su attempts. This type of correlation can later be used during alert investigation to distinguish expected laboratory activity from unexpected authentication failures.

Activity	Command / Source	Observed Result
Account verification	whoami	Current user identified as matthew
Failed authentication	su - matthew	Three controlled authentication failures
Primary log search	journalctl + authentication/PAM filters	pam_unix authentication failure and FAILED SU events
Recent event review	journalctl --since "5 minutes ago"	Recent authentication and su-related context

6. Security Relevance of the Logged Events

The journal records demonstrated that failed authentication attempts leave traceable operating-system evidence even when the login is unsuccessful. In this laboratory, the PAM authentication failure and FAILED SU messages exposed useful investigation fields such as the attempted account, UID, terminal session, and timestamp. These fields can be compared with future alerts to determine whether similar failures are expected administrative activity or require additional investigation.

7. Observed Authentication Log Fields

The failed events contained several fields that are useful during an investigation. The journal identified the Kali host, the matthew account, UID 1000, the pts/0 terminal session, the pam_unix authentication module, and explicit authentication failure / FAILED SU results. These details make it possible to connect a failed login event to a specific account, session, and point in time instead of reviewing an isolated error message.

Field	Observed Value	Investigation Use
Host	kali-mv	System that recorded the event
Account	matthew	Account used in the controlled authentication attempts
UID	1000	Local user identifier shown in the journal
Terminal	pts/0	Terminal session associated with the attempts

PAM context	pam_unix(su-l:auth)	Authentication component reporting the failure
Result	authentication failure / FAILED SU	Indicates that the authentication attempt was unsuccessful

8. Conclusions

The failed-authentication activity was successfully completed on the Kali Linux laboratory endpoint. Three safe authentication failures were generated against the local matthew account and the resulting events were located with journalctl. The PAM and FAILED SU records confirmed that the operating system logged the unsuccessful attempts and preserved useful context for investigation. The exercise demonstrated how authentication logs can be reviewed and correlated with known user activity without generating excessive login failures or interacting with systems outside the authorized laboratory environment.